
Physics I (SP015)

Topic 1: Physical Quantities and Measurements

1.1 Dimensions of Physical Quantities · 1.2 Scalars and Vectors

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1 What Is a Physical Quantity?

Before we do any physics, let us agree on one simple rule: every measurement must have a **number** *and* a **unit**.

Saying “the car is travelling at 90” tells us nothing useful. Is that $90 \frac{\text{km}}{\text{h}}$? $90 \frac{\text{m}}{\text{s}}$? The number alone is meaningless. A **physical quantity** is always a number together with its unit — for example, $\text{mass} = 5.0 \text{ kg}$, or $\text{time} = 3.0 \text{ s}$.

In this topic we learn two important ideas that help us work with physical quantities correctly:

- How to use **dimensions** to check whether equations make sense.
- How to tell apart **scalars** (quantities with magnitude only) from **vectors** (quantities with magnitude *and* direction).

2 Dimensions of Physical Quantities

Learning Outcomes

By the end of this section you should be able to:

- Define *dimension*.
- Determine dimensions of derived quantities.
- Verify the homogeneity of equations using dimensional analysis.

2.1 The Idea Behind Dimensions

Think about the quantity *speed*. You can measure speed in $\frac{\text{m}}{\text{s}}$, $\frac{\text{km}}{\text{h}}$, miles per hour, or even furlongs per fortnight! All of those are different *units*, but they are all measuring the same *type* of thing: how much **length** is covered per unit of **time**.

That underlying type — *length per time* — is what we call the **dimension** of speed. Dimensions tell us the *nature* of a quantity, not the specific unit used to measure it.

Definition 2.1: Dimension

Dimension refers to a way of expressing a physical quantity in terms of basic physical quantities (such as mass, length, time), regardless of the system of units used.

2.1.1 The Seven Base Dimensions

Every physical quantity — no matter how complicated — can be broken down into a combination of seven **base dimensions**:

Base Quantity	Dimension	SI Unit
Mass, m	M	kilogram (kg)
Length, l	L	metre (m)
Time, t	T	second (s)
Electric current, I	I	ampere (A)
Temperature, T	Θ	kelvin (K)
Amount of substance, n	N	mole (mol)
Luminous intensity, I_v	J	candela (cd)

2.1.2 How to Write Dimensions

We use square brackets to mean “the dimension of”. So:

$$[mass] = [m] = \text{M}, \quad [length] = [l] = \text{L}, \quad [time] = [t] = \text{T}$$

2.2 Finding Dimensions of Derived Quantities

Most quantities in physics are not base quantities — they are **derived quantities**, built from the base ones using a formula. To find the dimension of a derived quantity, just replace each symbol in its formula with its dimension and simplify.

The method in three steps:

- Write down the formula that defines the quantity.
- Replace every symbol with its dimension.
- Multiply or divide the dimensions, cancelling where possible.

Let us work through the most important ones.

Example 2.1: Velocity

Velocity is defined as displacement divided by time:

$$v = \frac{s}{t}$$

Replace s with L (it is a length) and t with T:

$$[v] = \frac{[s]}{[t]} = \frac{\text{L}}{\text{T}} = \text{L T}^{-1}$$

This just says: velocity is a *length* divided by a *time* — which makes perfect sense. Whether we write $\frac{\text{m}}{\text{s}}$ or $\frac{\text{km}}{\text{h}}$, the dimension stays L T^{-1} .

Example 2.2: Acceleration

Acceleration is the change in velocity per unit time:

$$a = \frac{\Delta v}{\Delta t}$$

We already know $[v] = \text{L T}^{-1}$, so:

$$[a] = \frac{[\Delta v]}{[\Delta t]} = \frac{\text{L T}^{-1}}{\text{T}} = \text{L T}^{-2}$$

Notice that dividing by time just adds another T^{-1} , giving T^{-2} overall.

Example 2.3: Force

Newton's second law tells us $F = ma$. Multiply the dimensions of mass and acceleration:

$$[F] = [m][a] = \text{M} \times \text{L T}^{-2} = \text{M L T}^{-2}$$

This is why $1 \text{ N} = 1 \text{ kg m/s}^2$.

Example 2.4: Work and Energy

Work is force times displacement ($W = Fs$):

$$[W] = [F][s] = \text{M L T}^{-2} \times \text{L} = \text{M L}^2 \text{T}^{-2}$$

All forms of energy — kinetic, potential, heat — share this same dimension $\text{M L}^2 \text{T}^{-2}$.

Example 2.5: Pressure

Pressure is force per unit area ($P = F/A$):

$$[P] = \frac{[F]}{[A]} = \frac{\text{M L T}^{-2}}{\text{L}^2} = \text{M L}^{-1} \text{T}^{-2}$$

One L on the top cancels with one from the L^2 on the bottom, leaving L^{-1} .

Here is a handy reference table of common SP015 quantities:

Table 1. Dimensions and SI units of common quantities in SP015.

Quantity	Symbol	Dimension	SI Unit
Displacement	s	L	m
Velocity	v	$L T^{-1}$	$\frac{m}{s}$
Acceleration	a	$L T^{-2}$	$\frac{m}{s^2}$
Force	F	$M L T^{-2}$	N
Work / Energy	W, E	$M L^2 T^{-2}$	J
Power	P	$M L^2 T^{-3}$	W
Pressure	P	$M L^{-1} T^{-2}$	Pa
Momentum	p	$M L T^{-1}$	$\frac{kg \cdot m}{s}$
Frequency	f	T^{-1}	Hz

2.3 Checking Equations with Dimensional Analysis

Here is one of the most powerful tricks in physics: you can use dimensions to check whether an equation *could* be correct — without doing any calculation at all.

Any valid equation must have the *same dimensions on both sides*, and every term being added must share the same dimension. This rule is called **dimensional homogeneity**.

Definition 2.2: Dimensional Homogeneity (not examinable)

An equation is **dimensionally homogeneous** if every term on both sides of the “=” sign has the same dimensions. If any term has a different dimension from the others, the equation is *definitely wrong*.

Key Concept: An Important Warning

Passing a dimensional check does **not** prove an equation is correct — it only tells you it *could* be correct.

For example, $v^2 = 2as$ and $v^2 = 7as$ are *both* dimensionally homogeneous, but only the first one is actually true. Dimensional analysis cannot detect wrong numerical factors like 2, π , or $\frac{1}{2}$ — those are dimensionless and invisible to this method.

How to check an equation:

- Work out the dimensions of *every separate term*.
- Check that all terms have the same dimensions.
- If they all match — the equation is dimensionally consistent. If any term differs — the equation is wrong.

Example 2.6: Checking $v^2 = u^2 + 2as$

We check each term one by one.

- **Term v^2 :**

$$[v^2] = (\text{L T}^{-1})^2 = \text{L}^2 \text{T}^{-2}$$

- **Term u^2 (same type of quantity as v):**

$$[u^2] = \text{L}^2 \text{T}^{-2}$$

- **Term $2as$ (the number 2 is dimensionless, so ignore it):**

$$[as] = \text{L T}^{-2} \times \text{L} = \text{L}^2 \text{T}^{-2}$$

All three terms give $\text{L}^2 \text{T}^{-2}$. ✓ The equation is **dimensionally homogeneous**.

Example 2.7: Catching a Wrong Equation

A student writes $F = mv$ (force = mass $\times$ velocity). Let us check whether this can possibly be right.

- **Left side:**

$$[F] = \text{M L T}^{-2}$$

- **Right side:**

$$[mv] = \text{M} \times \text{L T}^{-1} = \text{M L T}^{-1}$$

- **Comparison:**

$$\text{M L T}^{-2} \neq \text{M L T}^{-1}$$

The power of T is -2 on the left but only -1 on the right. They do not match, so the equation is **dimensionally wrong** — the student has written the formula for momentum ($p = mv$), not force.

*Pure numbers, angles (in radians), and ratios are all **dimensionless** — their dimension is just 1. You can always ignore them in a dimensional check.*

2.4 Common Mistakes

Watch Out! — Common Mistakes in Subtopic 1.1

- (a) **Forgetting to square the dimensions when squaring a quantity.**

$$[v^2] = (\text{L T}^{-1})^2 = \text{L}^2 \text{T}^{-2} \quad \text{not} \quad \text{L T}^{-2}$$

The power applies to *everything* inside the bracket.

- (b) **Thinking dimensionless numbers have dimensions.** Numbers like 2, π , or $\frac{1}{2}$ have no dimensions. Ignore them completely when checking an equation.
- (c) **Concluding an equation is correct just because it is dimensionally consistent.** A pass means the equation *might* be right. It does not mean it *is*

right.

- (d) **Trying to add quantities with different dimensions.** Writing $v + a$ makes no physical sense — LT^{-1} and LT^{-2} are different types of quantity.

2.5 Exercises

Exercise 2.1: Finding Dimensions

Find the dimension of each quantity using its defining formula:

- (a) Density ($\rho = m/V$)
- (b) Kinetic energy ($K = \frac{1}{2}mv^2$)
- (c) Pressure ($P = F/A$)
- (d) Momentum ($p = mv$)

Exercise 2.2: Checking Homogeneity

For each equation, state whether it is dimensionally homogeneous. Show all working.

- (a) $s = ut + \frac{1}{2}at^2$, where s is displacement, u is initial velocity, a is acceleration, t is time.
- (b) $P = \rho gh$, where P is pressure, ρ is density, g is acceleration due to gravity, h is depth.
- (c) $E = mva$, where E is energy, m is mass, v is velocity, a is acceleration.

Exercise 2.3: Spot the Error

A student claims that the kinetic energy of an object is given by $K = m^2v$. Without calculating anything, explain why this formula cannot be correct. What is the correct formula?

3 Scalars and Vectors

Learning Outcomes

By the end of this section you should be able to:

- (a) Define scalar and vector quantities.
- (b) Resolve vector into two perpendicular components (x and y axis)
- (c) Determine resultant of vectors. (remarks: limit to three vectors only)

3.1 What Is the Difference?

Not all physical quantities behave the same way. Some are completely described by a number and a unit. Others need a *direction* too.

A simple test: ask yourself — does direction matter for this quantity?

- *Temperature of a room is 30 °C.* Does direction matter? No — it is just a number. $\Rightarrow$ **scalar**.
- *A force of 50 N pushing to the right.* Does direction matter? Yes — push right and push left are completely different situations. $\Rightarrow$ **vector**.

Definition 3.1: Scalar

A **scalar** quantity has *magnitude only*. No direction is needed to describe it fully.

Definition 3.2: Vector

A **vector** quantity has both *magnitude* and *direction*. You must state both to describe it fully.

Table 2. Examples of scalar and vector quantities.

Scalar (magnitude only)	Vector (magnitude + direction)
Distance	Displacement
Speed	Velocity
Mass	Acceleration
Temperature	Force
Time	Momentum
Energy, Work	Weight
Power, Pressure	Torque

The most commonly confused pairs are: speed (scalar) vs. velocity (vector), and distance (scalar) vs. displacement (vector). Speed tells you how fast; velocity tells you how fast and in which direction.

3.2 How Do We Write and Draw Vectors?

In print: vectors are written in **bold**, e.g. $\mathbf{F}$, $\mathbf{v}$, $\mathbf{a}$.

In handwriting: put an arrow above the symbol, e.g. $\vec{F}$.

Magnitude: the magnitude (or size) of a vector $\mathbf{A}$ is written as $|\mathbf{A}|$ or simply A (non-bold italic).

A vector is drawn as an **arrow**. The length of the arrow shows the magnitude, and the direction the arrow points gives the direction of the vector.

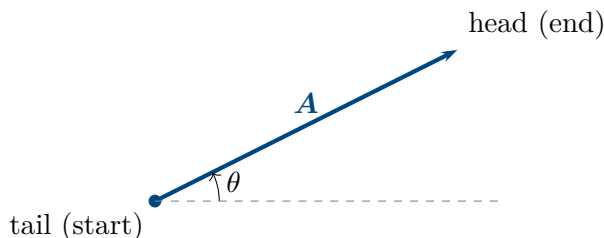


Figure 1. A vector $\mathbf{A}$ is drawn as an arrow from tail to head. Its magnitude is its length; its direction is the angle θ it makes with a reference line (usually the positive x -axis).

One more thing to remember: direction can be represented using **positive** and **negative** signs. For example, if we choose the rightward direction as positive, then the leftward direction is negative. Similarly, if upward is positive, then downward is negative.

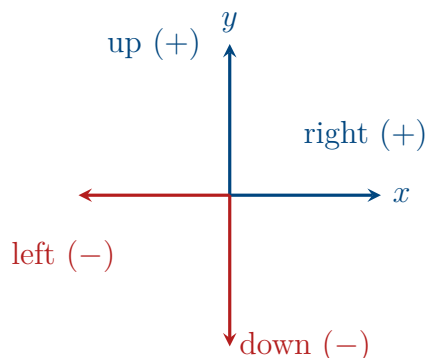
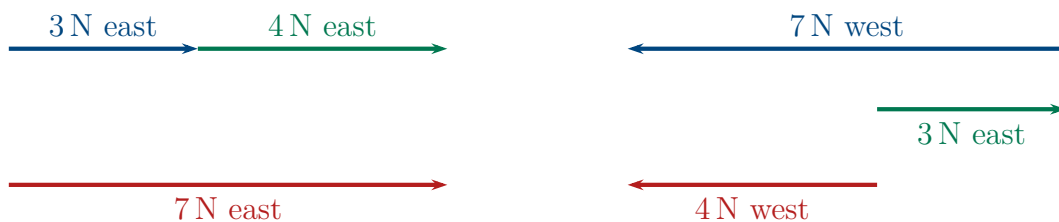


Figure 2. Positive and negative directions.

3.3 Adding Vectors

3.3.1 Adding Parallel Vectors

Adding vectors is easy when they act along the same straight line. For example, suppose we have to forces:



(a) Same direction: add magnitudes,
 $3\text{ N} + 4\text{ N} = 7\text{ N}$

(b) Opposite directions: subtract magnitudes,
 $3\text{ N} - 7\text{ N} = -4\text{ N}$

If the two forces act in the same direction, we simply add their magnitudes. If the vectors point in opposite direction, we subtract. In both cases the vectors are parallel, so the calculation is similar to ordinary addition and subtraction

The combined effect of two or more vectors is called the **resultant vector**.

3.3.2 When Vectors Are Not Parallel

Things become more interesting when the vectors do not point along the same line. Consider the following example:

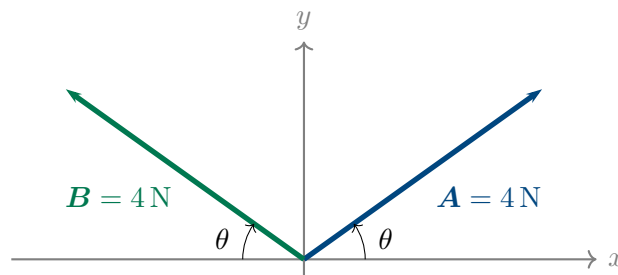


Figure 3. Two non-parallel vectors.

Can we simply add the numbers and say the answer is 8 N? No. The two vectors point in different directions, so we cannot use ordinary addition to combined them.

To find the resultant, we need a method that takes both magnitude and direction into account. This method is called **resolving**.

3.4 Resolving a Vector into Components

Any vector can be split into two perpendicular pieces — one along the x -axis (horizontal) and one along the y -axis (vertical). This splitting is called **resolution**.

Think of it this way: if you walk 5 m at 37° above the horizontal, you are effectively walking some distance east *and* some distance north at the same time. Resolution finds those two separate distances.

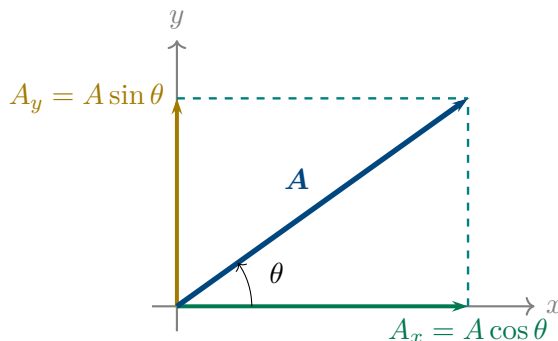


Figure 4. Resolving $\mathbf{A}$ into a horizontal component A_x and a vertical component A_y . The dashed lines form a right-angled triangle.

Definition 3.3: Vector Components

If a vector $\mathbf{A}$ has magnitude A and makes angle θ with the positive x -axis, its components are:

$$\begin{aligned} A_x &= A \cos \theta && (x\text{-component, horizontal}) \\ A_y &= A \sin \theta && (y\text{-component, vertical}) \end{aligned}$$

To *recover* the magnitude and angle from the components:

$$A = \sqrt{A_x^2 + A_y^2} \quad (\text{Pythagoras}) \quad (3.1)$$

$$\theta = \tan^{-1} \left| \frac{A_y}{A_x} \right| \quad (3.2)$$

*Components can be **positive or negative**. A component pointing in the negative x -direction is negative; one pointing upward (positive y) is positive. Always check which quadrant your vector lies in before assigning signs.*

3.5 Finding the Resultant — The Component Method

The component method is the standard calculation tool for adding vectors. It works for any number of vectors and any angles.

The five-step procedure:

- (a) Resolve *every* vector into x - and y -components.
- (b) Add all x -components together: $R_x = \sum A_x$
- (c) Add all y -components together: $R_y = \sum A_y$
- (d) Magnitude of resultant: $R = \sqrt{R_x^2 + R_y^2}$
- (e) Direction of resultant: $\theta = \tan^{-1} \left| \frac{R_y}{R_x} \right|$

Example 3.1: Two Perpendicular Forces

Two forces act on an object: $\mathbf{F}_1 = 10 \text{ N}$ (pointing right, 0°) and $\mathbf{F}_2 = 6 \text{ N}$ (pointing up, 90°). Find the resultant.

Step 1 — Components:

$$\begin{aligned} F_{1x} &= 10 \cos 0^\circ = 10 \text{ N}, & F_{1y} &= 10 \sin 0^\circ = 0 \text{ N} \\ F_{2x} &= 6 \cos 90^\circ = 0 \text{ N}, & F_{2y} &= 6 \sin 90^\circ = 6 \text{ N} \end{aligned}$$

Steps 2 & 3 — Sum:

$$F_x = 10 + 0 = 10 \text{ N}, \quad F_y = 0 + 6 = 6 \text{ N}$$

Step 4 — Magnitude:

$$F = \sqrt{10^2 + 6^2} = \sqrt{136} \approx 11.7 \text{ N}$$

Step 5 — Direction:

$$\theta = \tan^{-1}\left(\frac{6}{10}\right) \approx 31.0^\circ \text{ above the positive x-axis}$$

Answer:

$$F = 11.7 \text{ N at } 31.0^\circ \text{ above the positive x-axis}$$

Example 3.2: Two Vectors at a General Angle — Using a Table

Two displacements: $\mathbf{A} = 8 \text{ m}$ at 30° , and $\mathbf{B} = 5 \text{ m}$ at 120° . Find the resultant.

Step 1 — Organise in a table (this avoids careless errors):

Force	x -component / m	y -component / m
$\mathbf{A}$	$8 \cos 30^\circ = 6.93$	$8 \sin 30^\circ = 4.00$
$\mathbf{B}$	$5 \cos 60^\circ = -2.50$	$5 \sin 60^\circ = 4.33$
Total R	$6.93 + (-2.50) = 4.43$	$4.00 + 4.33 = 8.33$

Steps 4 & 5: Magnitude and direction of the resultant

$$R = \sqrt{4.43^2 + 8.33^2} \approx 9.43 \text{ m}, \quad \theta = \tan^{-1}\left(\frac{8.33}{4.43}\right) \approx 62.0^\circ$$

Answer:

$$R = 9.43 \text{ m at } 62.0^\circ \text{ above the positive x-axis}$$

Example 3.3: Three Forces — Using a Table

Three forces act on a body: $\mathbf{F}_1 = 12 \text{ N}$ at 210° , $\mathbf{F}_2 = 8 \text{ N}$ at 240° , and $\mathbf{F}_3 = 5 \text{ N}$ at 330° .

Step 1 — Organise in a table:

Force	x -component / N	y -component / N
$\mathbf{F}_1$	-10.39	-6.00
$\mathbf{F}_2$	-4.00	-6.93
$\mathbf{F}_3$	4.33	-2.50
Total F	-10.06	-15.43

Steps 4 & 5: Magnitude and direction of the resultant

$$F = \sqrt{(-10.06)^2 + (-15.43)^2} = 18.4 \text{ N}, \quad \theta = \tan^{-1}\left|\frac{-15.43}{-10.06}\right| = 56.9^\circ$$

Answer:

$$F = 18.4 \text{ N at } 56.9^\circ \text{ below the negative } x\text{-axis}$$

3.6 Common Mistakes

Watch Out! — Common Mistakes in Section 1.2

- (a) **Adding magnitudes directly.** $\mathbf{A} + \mathbf{B} \neq |\mathbf{A}| + |\mathbf{B}|$ unless both vectors point in exactly the same direction. Always use the component method.
- (b) **Mixing up sin and cos.** The rule $A_x = A \cos \theta$, $A_y = A \sin \theta$ *only* works when θ is measured from the positive x -axis. If θ is given from the y -axis, swap them.
- (c) **Dropping negative signs.** If a vector points left or downward, its component is *negative*. Forgetting this will give you a completely wrong answer. Always sketch the vector first.
- (d) **Writing a vector without its direction.** Saying “force = 10 N” is incomplete for a vector. You must also state its direction.

3.7 Exercises

Exercise 3.1: Scalar or Vector?

Classify each quantity. For each vector, state what information is missing if only a magnitude is given.

- (a) Room temperature: 25°C
- (b) A car’s velocity: $60 \frac{\text{km}}{\text{h}}$ due north
- (c) Mass of a book: 1.2 kg
- (d) A push on a door: 20 N
- (e) Energy stored in a battery: 100 J

Exercise 3.2: Resolving into Components

Find the x - and y -components of each vector.

- (a) 15 m at 40° above the positive x -axis
- (b) 20 N at 30° *below* the positive x -axis
- (c) $10 \frac{\text{m}}{\text{s}}$ directed straight upward
- (d) 25 N at 120° from the positive x -axis

Exercise 3.3: Resultant of Two Forces

Two forces act on a bolt: $\mathbf{F}_1 = 40 \text{ N}$ at 0° and $\mathbf{F}_2 = 30 \text{ N}$ at 90° .

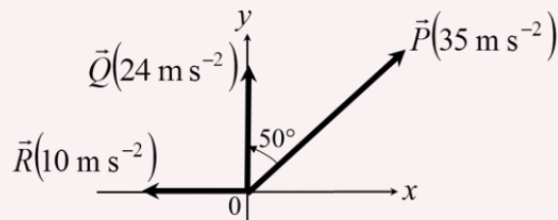
- (a) Find the x - and y -components of each force.
- (b) Find the magnitude and direction of the resultant.
- (c) Make a quick sketch to confirm your answer looks reasonable.

Exercise 3.4: Resultant of Two Vectors at a General Angle

Two displacement vectors: $\mathbf{d}_1 = 12\text{ m}$ at 25° and $\mathbf{d}_2 = 9\text{ m}$ at 70° . Find the magnitude and direction of the resultant displacement.

Exercise 3.5: Resultant of Three Vectors

- Given three vectors $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{R}$ as shown in the **FIGURE**. Use component method to calculate the resultant acceleration.



- At a picnic, there is a contest in which hoses are used to shoot water at a beach ball from three directions. As a result, three forces act on the ball, $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$. The magnitude of $\mathbf{F}_1$ and $\mathbf{F}_2$ are 50 N and 90 N respectively. Determine the magnitude of $\mathbf{F}_3$ and the angle θ such that the resultant force acting on the ball is zero.

